

Lab: Pattern Design Workshop — Estimating Yardage and Shaping

Objective

Practice the planning skills of crochet: yardage estimation, shaping calculation, construction method selection, and blocking prediction — all framed as prediction under uncertainty.

Materials Needed

- A garment or shaped pattern (hat, bag, or simple garment)
- Kitchen scale (for yarn weighing)
- Ruler and tape measure
- Calculator (or your mental math skills)
- Journal or notebook
- Blocking materials (spray bottle, pins, towel) for Exercise 4

Exercise 1: The Yardage Estimation Challenge (30 minutes)

Before starting your pattern, make three independent yardage estimates:

1. **Pattern estimate:** Use the pattern's stated yardage requirement
2. **Calculated estimate:** From your gauge, compute total stitches x yarn-per-stitch (measure 10 stitches, weigh the yarn used, extrapolate)
3. **Experience estimate:** Based on similar past projects, estimate from gut feeling

Method	Estimate (yards)	Confidence (1-5)
Pattern stated		
Calculated		
Experience-based		

After completing the project (or a significant portion), compare your estimates to actual usage. Which method was most accurate? What does this tell you about your model's yardage prediction capability?

Exercise 2: Shaping Calculation Workshop (30 minutes)

Plan the shaping for a simple hat from scratch:

1. Measure head circumference: ____ inches
2. Determine your gauge: ____ stitches per inch
3. Calculate starting stitch count: circumference x gauge = ____
4. Plan the crown decreases:
5. How many decrease rounds needed to close the crown?
6. How many stitches to decrease per round?
7. What crown shape will this produce (rounded vs. pointed)?

Draw a simple diagram showing stitch counts per round from brim to crown.

Bonus: Now plan a second version with a different crown shape. How does changing the decrease schedule change the prediction?

Exercise 3: Construction Method Comparison (20 minutes)

Choose a simple garment (vest, basic pullover) and outline two construction plans:

Feature	Top-Down Plan	Bottom-Up Plan
Starting point		
When can you check fit?		
How are armholes handled?		
How are pieces joined?		
Error correction ease		
Planning complexity		

Discuss: Which construction method gives you more opportunities for prediction error detection? Which requires more advance planning?

Exercise 4: Blocking Prediction Experiment (20 minutes)

If you have a completed swatch or small project:

1. Measure the pre-blocking dimensions
2. Predict the post-blocking dimensions (based on your knowledge of the fiber)
3. Block the piece (wet block or steam block as appropriate)
4. Measure the post-blocking dimensions

Measurement	Pre-Blocking	Predicted Post	Actual Post	Prediction Error
Width				
Length				
Stitch gauge				
Row gauge				

Reflect: How accurate were your blocking predictions? What did you learn about the hidden variables of post-construction transformation?

Discussion Questions for the Circle

1. How do you typically estimate yardage? Do you trust the pattern, calculate your own, or buy extra?
2. Have you ever run out of yarn on a project? What did you learn about your estimation model?
3. Do you prefer top-down or bottom-up construction? How does your preference relate to your planning style?

Wrap-Up

Write a brief reflection (3-5 sentences) on your planning skills. Where are your predictions most accurate (yardage, shaping, construction, blocking) and where do you have the most uncertainty? How would you improve your weakest planning area?